

Algebra in Action: Adventures with Anna

July 2026

1. The ‘A4’ lectures and notes

On 3rd September 2026, I will give a couple of lectures on the topic of ‘Algebra in Action: Adventures with Anna’ (‘A4’). The abstract is:

ABSTRACT. Across scientific domains, researchers seek to understand complex systems by decomposing them into simpler parts. These lectures will discuss how to build a theoretical and practical framework for such problems, using the tools and insights of metric algebraic geometry. I will also report on enjoyable conversations and collaborations to study real-world systems. The goal is to initiate a conversation on open questions and future directions.

These (evolving) notes complement the lectures. Their goal, for now, is to help prepare the audience. There will be three main sections: the past, the present, and the future.

For even more background, the lecture notes for (the first half of) my graduate course ‘Algebraic fundamental of representing data’ are available at <https://seigal.github.io/module1.pdf> and <https://seigal.github.io/module2.pdf>. Another short related reference is [6].

Comments on all are welcome!

2. Introduction

Algebra in action means algebra for mathematical data science: algebra devised or applied with the goal of answering a data analysis question. Examples of such questions may include those in biology, biomedicine, econometrics, literary studies, social sciences, etc. The questions have both trans-disciplinary and domain-specific aspects. We will reflect on what makes a data analysis question amenable, or not, to an algebraic point of view.

A central idea in the study of complex systems, such as those of the above applications, is the hypothesis of compositionality: observed variables (typically high-dimensional) can be described as combinations of (usually unobserved) simpler parts.

Formalized via random vectors, this hypothesis writes an observed random vector $\mathbf{x} \in \mathbb{R}^p$ as $\mathbf{x} = f(\mathbf{z})$ for a latent random vector $\mathbf{z} \in \mathbb{R}^q$, and a fixed unknown function $f : \mathbb{R}^q \rightarrow \mathbb{R}^p$. The special case where f is linear replaces it by a fixed, unknown matrix $A \in \mathbb{R}^{p \times q}$. We seek to recovering f or A and properties of $\mathbf{z}$ from samples of $\mathbf{x}$. This is a factorization problem. Many ideas in data analysis can be thought of as such factorization problems. We begin with the earliest, most classical examples.

3. The past

3.1. What is a factor? Multivariate data consists of a collection of vectors $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)}$ in $\mathbb{R}^p$. These each record measurements of p variables. Such data can be organized into a matrix $X \in \mathbb{R}^{n \times p}$: the n rows are the samples, the p columns index the variables, and entry X_{ij} is the value of variable j in sample i . For example:

Variable j	Sample i	X_{ij}
gene	cell	expression level of gene j in cell i
electrode	timepoint	voltage of electrode j at time i
pixel	image	intensity of pixel j in image i
word	document	count of word j in document i

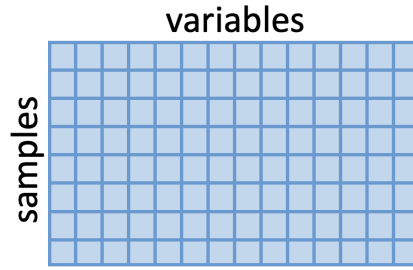


FIGURE 1. A data matrix, samples as rows, variables as columns

Data analysis seeks to uncover the building blocks that describe and explain such a matrix. Doing so amounts to fitting a model to the data. We view the vectors $\mathbf{x}^{(i)}, i = 1, \dots, n$ as samples of a vector of observed variables $\mathbf{x} = (x_1, \dots, x_p)^\top$. Writing observed variables as a linear combination of parts means imposing the model $\mathbf{x} = A\mathbf{z}$. We call $A \in \mathbb{R}^{p \times q}$ the mixing matrix: it encodes how the latent variables mix together to form the observed variables. At the level of samples, this says $\mathbf{x}^{(i)} = A\mathbf{z}^{(i)}$. Collecting together across all samples, this yields the matrix equation

$$X = ZA^\top,$$

where $X \in \mathbb{R}^{n \times p}$ is the data matrix and $Z \in \mathbb{R}^{n \times q}$ is the scores matrix. This product can equivalently be written as a sum

$$X = \sum_{k=1}^q \mathbf{z}_k \mathbf{a}_k^\top,$$

where $\mathbf{z}_k$ and $\mathbf{a}_k$ are the k th columns of Z and A , respectively.

The matrix product is rank q : the p observed variables have been expressed via q factors. By factor, here, we mean a process that describes the data. We don't get to see the latent factors. Instead, we extract (in the rows of Z) how each factor is expressed in a sample and (in the columns of A) how each variable participates in the factor. One thinks of this as *being* the factor: you are where you show up, or you are what you consist of. The collection of variables or samples that participate in a factor, e.g. those whose coefficient is above some threshold, is referred to as a module (not in the usual algebraic sense of module).

For example (in the table above): gene expression patterns can be encoded via gene modules, brain data via neural sources, images via features, and texts via their concepts and topics.

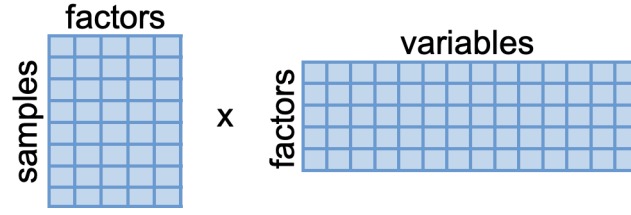


FIGURE 2. A low-rank decomposition $X = ZA^T$ of a data matrix writes it in terms of factors: the columns of Z record how much the factor is expressed in each sample, while the rows of A^T record how much each observed variable participates in it. Here there are $n = 8$ samples, $p = 14$ variables, and $q = 5$.

3.2. Principal component analysis (PCA). PCA is the first example of a model $\mathbf{x} = A\mathbf{z}$. The latent variables $\mathbf{z}$ in PCA are assumed to be uncorrelated. This means there are no linear relationships between them. The recovery of uncorrelated latent variables is intended to separate signals in the data. The mixing map A is assumed to be orthogonal: the transformation from $\mathbf{z}$ to $\mathbf{x}$ is an embedding that preserves distances.

DEFINITION 1 (Covariance). *Two variables z_1 and z_2 have covariance*

$$\text{Cov}(z_1, z_2) := \mathbb{E}[z_1 z_2] - \mathbb{E}[z_1]\mathbb{E}[z_2].$$

A collection of variables $\mathbf{z} = (z_1, \dots, z_q)$ has covariance matrix

$$\mathbb{E}[\mathbf{z} \otimes \mathbf{z}] - \mathbb{E}[\mathbf{z}] \otimes \mathbb{E}[\mathbf{z}].$$

This is the $q \times q$ PSD matrix with (i, j) entry $(\mathbb{E}[\mathbf{z} \otimes \mathbf{z}] - \mathbb{E}[\mathbf{z}] \otimes \mathbb{E}[\mathbf{z}])_{ij} = \mathbb{E}[z_i z_j] - \mathbb{E}[z_i]\mathbb{E}[z_j]$. The i th entry on the diagonal is the variance of z_i .

DEFINITION 2 (Uncorrelated). *Two variables z_1 and z_2 are uncorrelated if $\text{Cov}(z_1, z_2) = 0$. A collection of variables $\mathbf{z}$ are mutually uncorrelated if their covariance matrix $\Sigma_{\mathbf{z}}$ is diagonal.*

Applying the covariance to the relationship $\mathbf{x} = V\mathbf{z}$ yields

$$\Sigma_{\mathbf{x}} = V\Sigma_{\mathbf{z}}V^T,$$

where the matrices $\Sigma_{\mathbf{x}}$ and $\Sigma_{\mathbf{z}}$ are the covariance matrices of $\mathbf{x}$ and $\mathbf{z}$, respectively. The covariance $\Sigma_{\mathbf{z}}$ is diagonal, since the entries of $\mathbf{z}$ are uncorrelated. Now we observe that this factorization is the eigendecomposition. Its solution exists and is generically unique by the spectral theorem.

The matrix $\Sigma_{\mathbf{x}}$ is the population covariance of the observed variables. In practice, we do not have access to it and instead use the sample covariance matrix:

DEFINITION 3 (Sample covariance). *Given data $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)} \in \mathbb{R}^p$ with sample mean $\bar{\mathbf{x}} = \frac{1}{n} \sum_i \mathbf{x}^{(i)}$, the sample covariance matrix is*

$$S = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}^{(i)} - \bar{\mathbf{x}})(\mathbf{x}^{(i)} - \bar{\mathbf{x}})^T.$$

PCA computes the eigendecomposition of the sample covariance matrix, truncated to some rank r . That is, it keeps the top r eigenvalues (and their corresponding eigenvectors) and

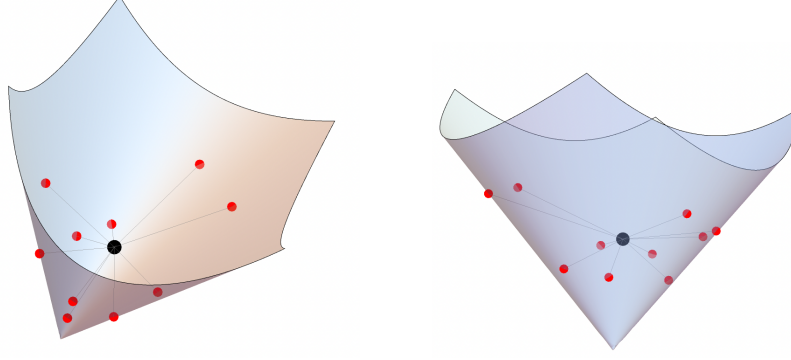


FIGURE 3. The 2×2 PSD cone (two views). Its boundary consists of matrices $\mathbf{v}\mathbf{v}^\top$ for $\mathbf{v} \in \mathbb{R}^2$. Here 10 mean-centered points (say $\mathbf{v}\mathbf{v}^\top$ for $\mathbf{v} \in \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)}\}$) on the boundary are shown in red. Their average (in black) is the sample covariance matrix of the data $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)}$.

sets the rest to 0. This is the best rank r approximation of the sample covariance matrix, by the Eckart-Young theorem.

Equivalently, PCA finds the singular value decomposition of the mean-centered data matrix X . That is, orthogonal matrices U and V and a diagonal matrix D such that $X = UDV^\top$. The two formulations are equivalent, because

$$\frac{1}{n}X^\top X = \frac{1}{n}(UDV^\top)^\top(UDV^\top) = VD^2V^\top.$$

3.3. Tensor decomposition. Above, our data consisted of points $\mathbf{x}^{(i)}$ in $\mathbb{R}^p$. Many datasets, though, carry multi-indexed structure, either on their space of variables, space of samples, or both. Examples include: fluorescence spectroscopy (samples $\times$ emission wavelengths $\times$ excitation wavelengths), color images (height $\times$ width $\times$ RGB channel), or neural recordings (neuron $\times$ experiment $\times$ time). See [5]. As with a data matrix, one can decompose it into rank one pieces. Now each piece is a factor. A factor is its expression in each sample, the amount each variable participates in it, and its presence across each context.

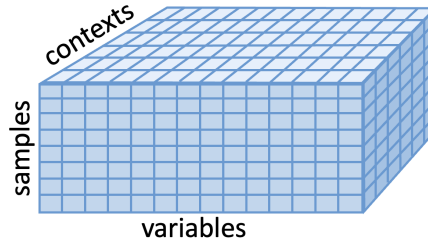


FIGURE 4. A data tensor, samples as rows, variables as columns, tubes as contexts.

Theoretical and practical questions arise for optimal extraction of interpretable factors from tensors of data (i.e. for low rank approximation of tensors). For example: how should decompositions of different ranks be compared or combined? For PCA/SVD, the solution as truncation of the eigendecomposition to the top r eigenvalues mean the optimal solutions have a natural nested structure. This doesn't happen in general for tensors.

3.4. Independent component analysis (ICA). Independent Component Analysis (ICA) makes two changes relative to PCA in the equation $\mathbf{x} = A\mathbf{z}$: it strengthens the uncorrelated assumption on $\mathbf{z}$ to independence, and it removes the orthogonality restriction on A .

DEFINITION 4. Variables $z_1, \dots, z_q$ are independent if their joint probability density function factors as $f_{\mathbf{z}}(t_1, \dots, t_q) = f_{z_1}(t_1) \cdots f_{z_q}(t_q)$.

Independence should be compared to the weaker notion of uncorrelated. Two variables being uncorrelated ruled out *linear* dependences between them; two variables being independent rules out *any* dependence between them: conditioning on the value of one variable doesn't change the density of the other.

The classical motivating picture for ICA is the *cocktail party problem*: p microphones each record a mixture $x_i = \sum_j A_{ij}z_j$ of p underlying, independent sound sources z_j ; the goal is to recover A and the sources from the microphone recordings alone. This is an example of a blind source separation problem. One central way to recover the parameters in an ICA model is via tensor decomposition of cumulant tensors [3], as we now describe.

DEFINITION 5 (Moments). The d -th moment tensor of $\mathbf{z}$ is $\mathcal{M}_d(\mathbf{z}) := \mathbb{E}[\mathbf{z}^{\otimes d}] \in S^d(\mathbb{R}^q)$ with entries $\mathcal{M}_d(\mathbf{z})_{i_1 \dots i_d} = \mathbb{E}[z_{i_1} \cdots z_{i_d}]$.

The first moment is the mean: $\mathcal{M}_1(\mathbf{z}) = \mu$; the second moment is the matrix $\mathcal{M}_2(\mathbf{z}) = \mathbb{E}[\mathbf{z} \otimes \mathbf{z}]$, which relates to the covariance by $\Sigma_{\mathbf{z}} = \mathcal{M}_2(\mathbf{z}) - \mu \otimes \mu$, as we saw above.

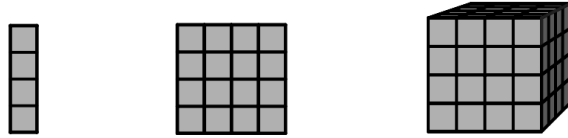


FIGURE 5. The first moment is a vector, the second is a matrix, the third is a third order tensor.

DEFINITION 6 (Moments and cumulants). For $\mathbf{t} = (t_1, \dots, t_q)^T$, the moment generating function (mgf) of $\mathbf{z}$ is $M_{\mathbf{z}}(\mathbf{t}) = \mathbb{E}[\exp\langle \mathbf{t}, \mathbf{z} \rangle]$, and the cumulant generating function (cgf) is $K_{\mathbf{z}}(\mathbf{t}) = \log M_{\mathbf{z}}(\mathbf{t})$. We have

$$M_{\mathbf{z}}(\mathbf{t}) = \sum_{d \geq 0} \frac{1}{d!} \langle \mathbf{t}^{\otimes d}, \mathcal{M}_d(\mathbf{z}) \rangle.$$

and define the d -th cumulant tensor $\mathcal{K}_d(\mathbf{z}) \in S^d(\mathbb{R}^q)$ as the analogous coefficient of $K_{\mathbf{z}}$:

$$K_{\mathbf{z}}(\mathbf{t}) = \sum_{d \geq 1} \frac{1}{d!} \langle \mathbf{t}^{\otimes d}, \mathcal{K}_d(\mathbf{z}) \rangle.$$

PROPOSITION 7 (The cumulant tensors of independent variables are diagonal). If $z_1, \dots, z_q$ are independent, $\mathcal{K}_d(\mathbf{z})$ is diagonal for every d .

PROOF. Independence gives $M_{\mathbf{z}}(\mathbf{t}) = \mathbb{E}[\exp(t_1 z_1) \cdots \exp(t_q z_q)] = \prod_{i=1}^q \mathbb{E}[\exp(t_i z_i)]$, so

$$K_{\mathbf{z}}(\mathbf{t}) = \log M_{\mathbf{z}}(\mathbf{t}) = \sum_{i=1}^q \log \mathbb{E}[\exp(t_i z_i)],$$

a sum of univariate functions. Hence the degree- d coefficient tensor $\mathcal{K}_d(\mathbf{z})$ has entries supported only on indices $(i, i, \dots, i)$. $\square$

DEFINITION 8 (The action of A on a symmetric tensor). For $A \in \mathbb{R}^{p \times q}$ and $S \in S^d(\mathbb{R}^q)$, define $A \bullet S \in S^d(\mathbb{R}^p)$ by

$$(A \bullet S)_{i_1 \dots i_d} = \sum_{\alpha_1, \dots, \alpha_d=1}^q S_{\alpha_1 \dots \alpha_d} A_{i_1 \alpha_1} \cdots A_{i_d \alpha_d}.$$

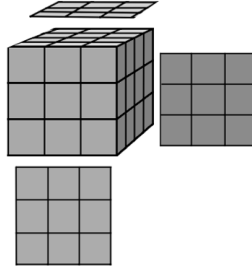


FIGURE 6. The action $A \bullet S$ multiplies a tensor by a matrix along each index.

For $d = 2$, $(A \bullet S)_{ij} = \sum_{\alpha\beta} S_{\alpha\beta} A_{i\alpha} A_{j\beta} = (ASA^T)_{ij}$: this recovers the congruence action. The expression above is a product on the left and a sum on the right. For a matrix, we saw that a factorization can be viewed as a product or as a sum, too.

PROPOSITION 9 (Multilinearity). Let $\mathbf{x} = A\mathbf{z}$ for $A \in \mathbb{R}^{p \times q}$. Then $\mathcal{M}_d(\mathbf{x}) = A \bullet \mathcal{M}_d(\mathbf{z})$ and $\mathcal{K}_d(\mathbf{x}) = A \bullet \mathcal{K}_d(\mathbf{z})$.

PROOF.

$$\mathcal{M}_d(\mathbf{x})_{i_1 \dots i_d} = \mathbb{E}[x_{i_1} \cdots x_{i_d}] = \mathbb{E}\left[\prod_{k=1}^d \sum_{\alpha_k} A_{i_k \alpha_k} z_{\alpha_k}\right] = \sum_{\alpha_1, \dots, \alpha_d} A_{i_1 \alpha_1} \cdots A_{i_d \alpha_d} \mathbb{E}[z_{\alpha_1} \cdots z_{\alpha_d}],$$

using linearity of expectation. The last factor is $\mathcal{M}_d(\mathbf{z})_{\alpha_1 \dots \alpha_d}$, giving $(A \bullet \mathcal{M}_d(\mathbf{z}))_{i_1 \dots i_d}$. The same proof applies to the coefficient tensors from the cumulant generating function. $\square$

PROPOSITION 10. Fix $d \geq 3$. Writing $A = [\mathbf{a}_1 \cdots \mathbf{a}_p]$ (columns) and $\lambda_i := \mathcal{K}_d(\mathbf{z})_{i, i, \dots, i}$ (the i -th source's order- d cumulant),

$$\mathcal{K}_d(\mathbf{x}) = A \bullet \mathcal{K}_d(\mathbf{z}) = \sum_{i=1}^p \lambda_i \mathbf{a}_i^{\otimes d}.$$

PROOF. The cumulant tensors of mutually independent variables are diagonal. By the cumulant analog of multilinearity of moments (Proposition 9), $\mathcal{K}_d(\mathbf{x}) = A \bullet \mathcal{K}_d(\mathbf{z}) = \sum_i \lambda_i (A \bullet \mathbf{e}_i^{\otimes d})$. We have $(A \bullet \mathbf{e}_i^{\otimes d})_{j_1 \dots j_d} = A_{j_1 i} \cdots A_{j_d i} = (\mathbf{a}_i^{\otimes d})_{j_1 \dots j_d}$, so $A \bullet \mathbf{e}_i^{\otimes d} = \mathbf{a}_i^{\otimes d}$. $\square$

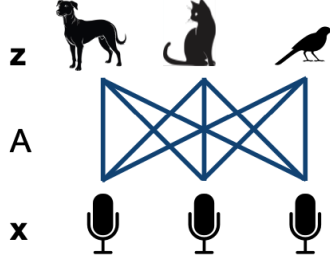


FIGURE 7. ICA for the cocktail party problem. Sources $\mathbf{z}$ are mixed together via mixing matrix A to produce observed variables $\mathbf{x}$.

Methods to analyze data give factorizations. In this section, we saw two examples: PCA and the eigendecomposition, and ICA and symmetric tensors decomposition. A third classical example is the connection between LDL decompositions (or the Cholesky or QR decompositions) and linear structural equation model, but we don't have time for that here.

4. The present

This section looks at present-day data analysis methods and the factorizations that underpin them. In the lectures, we'll discuss collaborations to use these factorizations to tackle data analysis questions. There are two themes:

- (1) How systems change across contexts: so far we have looked at a single set of observed variables $\mathbf{x} = (x_1, \dots, x_p)$. Many situations call for a set of variables $\mathbf{x}$ observed under multiple contexts (also called views or environments), such as patients, batches, time periods, etc.
- (2) Identification under weaker assumptions: the factorizations we have seen so far are identifiable, up to benign and unavoidable sources of non-identifiability, such as the ordering of the recovered factors. How much can the assumptions we make (such as independence) be relaxed while still yielding an identifiable factorization?

4.1. Multi-context factorizations. The idea of principal components that are shared across multiple datasets originates in work of Flury in the 1980s [4]. The *hypothesis of common principal components* says that the covariance matrices in the contexts share the same principal components but their weights vary: they have the same eigenvectors but different eigenvalues:

$$(1) \quad \Sigma_j = V D_j V^T,$$

where $V \in \mathbb{R}^{p \times p}$ is the matrix of eigenvectors and $D_j \in \mathbb{R}^{p \times p}$ the corresponding eigenvalues in context j . In practice, as in usual PCA, the eigendecompositions are truncated to the top r , to give $V \in \mathbb{R}^{p \times r}$ and $D_j \in \mathbb{R}^{r \times r}$.

This replaces diagonalization of a covariance with simultaneous diagonalization. From an algorithmic point of view, it began a line of joint/simultaneous diagonalization methods.

The link between simultaneous diagonalization and tensor decomposition is well known. In this example, stacking the k covariance matrices builds a $p \times p \times k$ tensor T with $T_{\alpha\beta j} = (\Sigma_j)_{\alpha\beta}$; see below. Decomposition of this tensor is a simultaneous diagonalization problem.

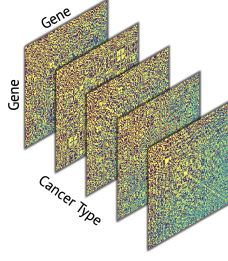


FIGURE 8. A stack of $p \times p$ covariance matrices, one for each of k contexts, forms a $p \times p \times k$ tensor. In this picture, $p = 500$ and $k = 5$.

PROPOSITION 11. *The partially symmetric tensor decomposition*

$$(2) \quad T = \sum_{i=1}^r \mathbf{v}_i^{\otimes 2} \otimes \mathbf{d}_i,$$

is equivalent to the coupled matrix decompositions in (1), where $\mathbf{v}_i \in \mathbb{R}^p$ are the columns of V and where $\mathbf{d}_i = ((D_1)_{ii}, \dots, (D_k)_{ii}) \in \mathbb{R}^k$.

PROOF. The decomposition in (2) says $T_{\alpha\beta j} = \sum_{i=1}^r (\mathbf{v}_i)_\alpha (\mathbf{v}_i)_\beta (\mathbf{d}_i)_j = \sum_{i=1}^r V_{i\alpha} V_{i\beta} (D_j)_{ii} = (VD_j V^\top)_{\alpha,\beta}$, so $T_{:, :, j} = VD_j V^\top$. $\square$

For r small enough, this decomposition is identifiable: the factors $\mathbf{v}_i$ and $\mathbf{d}_i$ are unique up to scale and permutation, see e.g. [2]. Orthogonality of the $\mathbf{v}_i$ is not required for identifiability. Our recent work on multi-context PCA [7] is a new algorithm for partially symmetric tensor decomposition to find axes of variation in data and the combinations of contexts in which they appear.

The case of two contexts has been of particular interest. The two contexts then represent a foreground datasets (such as a disease group) and a background dataset (such as healthy controls). The goal is to identify structure present in a foreground dataset after removing the effect of the background. Several methods have been devised for this purpose including contrastive PCA [1] and contrastive ICA [8].

4.2. New decompositions.

- latent variables
- higher-order dependence

5. The future

Coming soon :-)

References

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